Supplementary Information

USCrash: A Harmonized U.S. Nationwide Traffic Crash Dataset

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Supplementary Note 1. Data sources and coverage

USCrash contains 65,541,414 crash-level records in 511 annual files from 41 states and the District of Columbia, spanning 2001–2026. Table S1 lists the retained years, record counts, acquisition methods, and source channels. Data were obtained through the four approaches described in the main text: direct data downloading (D), public data services/APIs (S), web-based query and export systems (E), and formal agency requests (R). Multiple methods are listed where the source channel changed across years or an agency file supplemented a public release.

Several source changes are relevant to interpretation. Florida uses FDOT GIS records for 2007–2019 and Signal Four Analytics tables for 2020–2025. Michigan combines historical GIS records with subsequent Traffic Crash Facts exports. Minnesota combines MnCrash with Road Safety Information Center detail. New Jersey uses annual linked tables and dashboard releases for different years. Nevada uses the public 2019 extract and the agency-provided 2020–2024 tables. Washington combines WSP CAT records with agency-supplied SRFF detail for 2020–2026. Ohio combines TIMS with the linked research archive at Figshare record 29437694.

Table S1: Data sources and retained coverage of USCrash. Source names link to the corresponding agency or release channel.

State	Years	Files	Records	Method	Source
Alaska (AK)	2021–2025	5	42,985	S	DOT&PF public crash portal
Alabama (AL)	2014–2025	12	1,660,326	D, S	CARE / University of Alabama CAPS
Arkansas (AR)	2021–2025	5	401,279	S	ARDOT ACAT
Arizona (AZ)	2020–2025	6	707,288	R	ADOT ACIS export
California (CA)	2016–2026	11	4,670,092	D	CHP California Crash Reporting System
Colorado (CO)	2007–2026	20	2,097,474	D	CDOT annual crash listings
Connecticut (CT)	2015–2026	12	1,215,746	S	CTDOT / UConn crash repository
District of Columbia (DC)	2010–2026	17	339,580	S	Open Data DC / DDOT
Delaware (DE)	2009–2026	18	588,483	S	Delaware Open Data / DelDOT
Florida (FL)	2007–2025	19	6,048,502	D, S	FDOT State Safety Office; Signal Four Analytics
Georgia (GA)	2021–2025	5	1,895,149	E	GDOT crash dashboard
Iowa (IA)	2015–2025	11	597,522	S	Iowa DOT crash data
Idaho (ID)	2005–2023	19	447,400	S	ITD public crash services
Illinois (IL)	2014–2025	12	3,625,428	S	IDOT annual crash extracts
Kansas (KS)	2016–2026	11	603,643	S	KDOT public crash services
Kentucky (KY)	2020–2026	7	907,450	D	KYTC / KSP collision extracts
Massachusetts (MA)	2001–2026	26	3,305,764	S	MassDOT IMPACT
Maine (ME)	2020–2026	7	219,143	E	Maine Public Crash Query Tool
Michigan (MI)	2017–2025	9	2,660,935	S, E	Michigan State Police / Traffic Crash Facts
Minnesota (MN)	2016–2026	11	747,224	S, E	MnCrash; Road Safety Information Center
Missouri (MO)	2002–2003, 2013–2026	16	1,968,862	S	Missouri State Highway Patrol STARS
Mississippi (MS)	2004–2025	22	1,578,591	D	MDOT ReportBeam / eCrash
Montana (MT)	2016–2025	10	219,482	D	Montana DOT crash database
North Carolina (NC)	2016–2026	11	1,746,541	S	NCDOT public crash tables
Nebraska (NE)	2010–2026	17	675,175	E	Nebraska DOT NTIP
New Jersey (NJ)	2001–2026	26	7,271,660	D, S	NJDOT raw crash files; AASHTOWare dashboard
New Mexico (NM)	2019	1	23,896	S	NMDOT public GIS service
Nevada (NV)	2019–2024	6	347,123	S, R	Nevada DOT open service and agency extract
New York (NY)	2021–2024	4	1,525,123	S	New York State DMV Case Information

Table S1 (continued)

State	Years	Files	Records	Method	Source
Ohio (OH)	2017–2025	9	2,314,530	D, S	ODOT TIMS; linked research archive
Oklahoma (OK)	2019–2021	3	14,055	S	Oklahoma Highway Safety Office KAB layer
Oregon (OR)	2007–2024	18	867,941	D	Oregon DOT annual crash files
Pennsylvania (PA)	2006–2025	20	2,436,722	D	PennDOT statewide crash files
South Dakota (SD)	2011–2026	16	281,142	D, S	South Dakota DOT crash files and GIS
Tennessee (TN)	2016–2026	11	2,379,533	D	Tennessee TITAN annual extract
Texas (TX)	2019–2026	8	4,166,727	R	TxDOT CRIS agency extract
Utah (UT)	2018–2026	9	519,400	S	UDOT crash locations
Virginia (VA)	2009–2026	18	2,139,331	S	Virginia DOT / DMV crash services
Vermont (VT)	2010–2026	17	181,582	S	VTrans Crash Public Query Tool
Washington (WA)	2017–2026	10	1,168,229	E, R	WSP CAT; agency-supplied SRFF files
Wisconsin (WI)	2017–2022	6	801,193	D	WisDOT DT4000 annual archives
Wyoming (WY)	2016–2025	10	133,163	R	WYDOT agency-supplied crash tables
Total	**2001–2026**	**511**	**65,541,414**		

Supplementary Note 2. Crash-level preparation and quality control

Preparation produces one retained row per crash within the applicable source population. Where information is distributed across linked tables, the state-specific scripts use source crash identifiers and entity keys to summarize supported counts and involvement indicators. Repeated observations are reconciled under the source-specific identifier rules. Table S2 summarizes the preparation used in the current dataset.

Quality control checks input, excluded, duplicate, and retained counts; the 77-column structure; configured identifier uniqueness; date and time representations; category validity; and severity/count consistency. Coordinate checks examine numeric validity and configured state bounds. A separate geographic screen uses buffered state boundaries for candidate locations. Month-only Pennsylvania records retain year and month without an invented day. New York records use documented row-based surrogate identifiers where the public release omits a stable crash key.

Coordinate supplementation uses additional official sources only where the source linkage and geographic checks support a match. Exact identifiers and date/time checks are used where available; reviewed composite keys are used for selected California sources. Conflicting candidates are withheld. Coordinate changes and their source references are recorded in the revision ledgers. These steps supplement locations without treating the donor records as additional crashes.

Table S2: Source-specific crash-level preparation. The table summarizes transformations used by the current dataset; implementation details are supplied with the code.

State(s)	Preparation
CA	Link crash, party, and injured-person records by collision identifier; derive supported injury severity, counts, and road-user involvement.
FL	Retain CODEABLE=Y records from Signal Four Analytics for 2020–2025. Do not map the older source most-harmful-event field to first harmful event.
KY	Link incident and person records; use the most severe supported person-injury category and documented incident-level fallbacks.
MN	Match MnCrash and Road Safety Information Center records using the documented crash identifier and date checks; retain supported detailed attributes.
MT	Aggregate supported crash, vehicle, and person information. Transform documented state-plane coordinates to WGS84.

Table S2 (continued)

State(s)	Preparation
NJ	Prepare annual crash records from linked accident, driver, occupant, pedestrian, and vehicle tables, or the applicable dashboard release.
NV	Combine the agency crash, person, and vehicle extract; resolve duplicate crash identifiers using the documented severity/completeness/source-row ordering.
OH	Prepare the linked research tables and TIMS crash-location records using version-specific inputs; distinguish detail available in different years.
PA	Join the crash and flags tables by crash identifier; preserve reported year/month without constructing an unreported date.
SD	Prepare crash-level attributes from the applicable detailed or public GIS release, using linked entities where supplied.
TN	Join collision, unit, and person tables; retain agreed unit-level context and leave conflicting unit values unresolved.
TX	Prepare supported crash attributes from the agency CRIS extract using the reviewed field selection and source definitions.
UT, VA	Normalize the applicable public crash-level schema and retain documented source identifiers and context.
VT	Prepare crash-level fields and reconcile repeated source crash observations under the configured rule.
WA	Reconcile CAT records with the agency-supplied SRFF files by report number, including cross-year overlap checks; retain CAT coverage and supported SRFF detail.
WY	Split the multiyear crash, vehicle, and involved-person package into annual source tables; harmonize the crash-level file.
Other states	Use the configured crash-level source directly, with source-specific parsing, inclusion and identifier rules.

Supplementary Note 3. MMUCC-guided semantic harmonization rules

The common schema contains 77 columns in eight groups: identifiers and provenance (7), time (7), location (5), severity and outcomes (6), vehicles and road users (13), collision and event characteristics (10), environment conditions (12), and roadway and behavior context (17). Table S3 defines the columns; Table S4 summarizes the main harmonization rules. The field names below are the exact names used in the data files.

MMUCC provides the semantic reference for the common attributes. State dictionaries and reporting definitions determine whether a source value supports a particular mapping. Raw-value and provenance columns retain the basis of the mapped attributes. Missing information is not replaced by an assumed date, coordinate, injury level, or involvement indicator. A blank value therefore differs from an explicitly reported negative or unknown value.

Table S3: The 77 MMUCC-guided semantic harmonized schema. Column counts include raw-value and provenance fields. Blank values and precision/provenance fields are interpreted together.

Field	Type	Definition
Identifiers and provenance (7 columns)		
`crash_uid`	string	Unique crash identifier within the assembled dataset.
`state`	string	Two-letter state code; DC denotes the District of Columbia.
`source_crash_id`	string	Source crash identifier, or documented deterministic surrogate.
`source_schema_id`	string	Identifier of the source schema and applicable configuration.
`source_file`	string	Configured input-file path relative to the state-source directory.
`source_row_number`	integer	One-based input data-row number, excluding the header.
`pipeline_version`	string	Processing version and applicable revision identifier.
Time (7 columns)		
`crash_date`	date	Reported crash date, formatted as YYYY-MM-DD when available.
`crash_year`	integer	Reported year or year derived from the crash date.
`crash_month`	integer	Reported month or month derived from the crash date.
`date_precision`	string	Available date precision, such as day or month.
`crash_time`	time	Crash clock time formatted as HH:MM:SS.
`crash_datetime`	datetime	Combined crash date and clock time when both are available.
`time_precision`	string	Available clock-time precision, including hour or reported interval.
Location (5 columns)		
`latitude`	float	Available WGS84 latitude, in decimal degrees.
`longitude`	float	Available WGS84 longitude, in decimal degrees.
`coordinate_status`	string	Coordinate validity or missingness classification.
`county`	string	County name or code retained from the source.
`city`	string	City name or code retained from the source.
Severity and outcomes (6 columns)		
`severity_kabco`	string	Crash injury severity: K, A, B, C, O, or U.
`severity_raw_code`	string	Source severity code used for mapping.
`severity_raw_label`	string	Source severity label used for mapping.
`severity_provenance`	string	Basis of the severity mapping or derivation.
`fatalities`	integer	Number of fatalities reported or supported by linked records.
`injuries`	integer	Number of injuries as supported by the source definition.
Vehicles and road users (13 columns)		
`vehicles`	integer	Exact involved-vehicle count, where supported.
`vehicles_minimum`	integer	Exact count or lower bound for a grouped vehicle count.
`vehicles_raw`	string	Source vehicle-count value, including grouped values.
`vehicles_precision`	string	Whether the vehicle count is exact, a lower bound, or unavailable.
`pedestrian_involved`	string	Indicator of pedestrian involvement.
`pedestrian_count`	integer	Number of involved pedestrians, where reported.
`bicyclist_involved`	string	Indicator of bicyclist involvement.
`bicyclist_count`	integer	Number of involved bicyclists or source-defined pedalcyclists.
`motorcycle_involved`	string	Indicator of motorcycle involvement.
`motorcycle_count`	integer	Number of involved motorcycles, where reported.

Table S3 (continued)

Field	Type	Definition
commercial_vehicle_involved	string	Indicator of source-defined commercial-vehicle involvement.
commercial_vehicle_count	integer	Number of involved commercial vehicles, where reported.
school_bus_related	string	Indicator of school-bus-related involvement.
Collision and event characteristics (10 columns)		
manner_of_collision_mmucc	string	MMUCC-guided collision-manner category.
manner_raw_code	string	Source collision-manner code.
manner_raw_label	string	Source collision-manner label.
manner_provenance	string	Basis of the collision-manner mapping.
first_harmful_event	string	First harmful event when the source identifies this concept.
first_harmful_event_group	string	Broader analytical grouping of the first harmful event.
first_harmful_event_raw_code	string	Source event code used for mapping.
first_harmful_event_raw_label	string	Source event label used for mapping.
first_harmful_event_provenance	string	Basis of the first-harmful-event mapping.
collision_partner_group	string	Collision counterpart identified by the supported event mapping.
Environment conditions (12 columns)		
weather	string	Harmonized weather condition.
weather_raw_code	string	Source weather code.
weather_raw_label	string	Source weather label.
weather_provenance	string	Basis of the weather mapping.
light_condition	string	Harmonized lighting condition.
light_raw_code	string	Source lighting code.
light_raw_label	string	Source lighting label.
light_provenance	string	Basis of the lighting mapping.
road_surface_condition	string	Harmonized road-surface condition.
road_surface_raw_code	string	Source road-surface code.
road_surface_raw_label	string	Source road-surface label.
road_surface_provenance	string	Basis of the road-surface mapping.
Roadway and behavior context (17 columns)		
functional_class	string	Roadway functional class, with standardization recorded by provenance.
functional_class_raw_code	string	Source functional-class code.
functional_class_raw_label	string	Source functional-class label.
functional_class_provenance	string	Whether the class is harmonized or remains source-specific.
intersection_related	string	Source-supported relationship to an intersection.
intersection_type	string	Intersection type, retaining source-specific detail where necessary.
work_zone_related	string	Indicator or category of work-zone involvement.
work_zone_type	string	Source-supported work-zone type or location.
alcohol_involved	string	Indicator of source-reported alcohol involvement.
drug_involved	string	Indicator of source-reported drug involvement.
speeding_involved	string	Indicator of source-reported speeding involvement.
speeding_provenance	string	Basis of the speeding indicator.
hit_and_run	string	Indicator of hit-and-run involvement.
urban_rural	string	Source-supported urban or rural classification.
posted_speed_limit	float	Posted speed limit in miles per hour.
relation_to_trafficway	string	Location relative to the trafficway, where reported.
number_of_lanes	integer	Reported roadway lane count.

Configured source-field bindings are provided in `tables/state_field_mappings.csv`, with state, schema version, applicable years, and source/target field names. Reviewed category mappings are compiled in `tables/state_value_crosswalks.csv`, including the originating crosswalk file and mapping provenance. The corresponding configurations are retained in `tables/state_mapping_rules.json`. Entity-level derivations and conditional preparation steps remain defined by the accompanying scripts.

Table S4: Key MMUCC-guided semantic harmonization rules. Configured field bindings and categorical mappings are provided in the accompanying CSV files described above.

Attribute	**Rule**
Identifiers and provenance	Use a source identifier within its documented state/year scope. If no stable source identifier is released, retain a deterministic row-based surrogate and its source-file reference.
Date and time	Standardize reported dates and clock times while retaining precision. Month-only dates remain month-only; hour-coded times do not acquire minute-level precision.
Location	Express supported coordinates in WGS84. Transform projected coordinates only when the source coordinate system is documented; retain source county and city labels.
Severity and outcomes	Map documented injury categories to KABCO: fatal (K), serious injury (A), minor injury (B), possible injury (C), no injury (O), and unknown/unresolved (U). Generic injury is not assigned an unsupported A/B/C level.
Vehicle counts	Separate exact counts from grouped counts. For example, a source value of three or more gives a lower bound of three, not an exact count of three.
Road-user involvement	Use reported crash-level indicators or supported linked-record counts. An absent field does not establish that a road user was absent.
Collision and event characteristics	Keep collision manner, first harmful event, and collision counterpart separate. A most harmful event is not substituted for a first harmful event.
Motorcycles	Motorcycles are motor vehicles. Motorcycle involvement is not automatically a subtype of crashes not involving collision with a motor vehicle in transport.
Environment conditions	Map weather, lighting, and surface categories using the source dictionary; preserve source codes/labels and any combined distinctions.
Roadway and behavior context	Retain source-supported categories and indicators. Functional class or other context that cannot be standardized retains source-specific provenance.
Missing and unknown values	Leave unavailable or unmappable attributes blank or explicitly unknown as specified by the mapping. Preserve reported unknown and not-applicable categories where supported.
Mapping provenance	Record the source code/label and mapping basis for attributes with provenance fields. State- and version-specific configurations define the applicable rules.

Supplementary Note 4. Official-count validation and benchmark comparison

The expanded benchmark uses calendar year 2021 and comprises 38 cities and two Florida counties: the 20 areas in the main text and 20 additional cities. The additional sample includes Idaho, Kansas, and South Dakota, extending the comparison to 17 states. Official counts come from annual reports, published tables, or official query systems. USCrash selections follow the corresponding source municipality or county definition; the three Idaho cities use valid coordinates within Census TIGER 2021 municipal boundaries because source city labels are unavailable. The Texas selections additionally use the official reportable flag and agency city identifier; Phoenix uses the official raw city code intersected with the retained crash identifiers. Philadelphia County is coterminous with Philadelphia city. Orange and Hillsborough are county comparisons, not the municipal totals of Orlando and Tampa.

US-Accidents counts use the March 2023 release, selecting records whose start time falls in 2021 and whose start coordinate intersects the corresponding Census TIGER 2021 city or county boundary. For dataset count N and official reference N_{official}, the reported difference is

$$\text{Difference (\%)} = 100\,\frac{N - N_{\text{official}}}{N_{\text{official}}}.$$

The aggregate difference is calculated from the summed counts rather than the mean of the area percentages. Table S5 gives the comparison counts and source references.

Table S5: Expanded official-count benchmark for 38 cities and two Florida counties in 2021. Starred cities extend the main-text sample. Differences use the official count as denominator; area names link to official sources.

Areas 1–20						Areas 21–40					
Area	Official	USCrash	Diff.	US-Accidents	Diff.	Area	Official	USCrash	Diff.	US-Accidents	Diff.
Phoenix, AZ	40,833	40,876	+0.11%	9,348	-77.11%	Wichita, KS*	7,379	7,379	0.00%	426	-94.23%
Los Angeles, CA	29,953	30,033	+0.27%	33,742	+12.65%	Boston, MA	4,139	4,139	0.00%	3,207	-22.52%
San Diego, CA	8,338	8,332	-0.07%	11,014	+32.09%	Worcester, MA*	4,722	4,722	0.00%	92	-98.05%
San Francisco, CA	5,805	5,767	-0.65%	2,929	-49.54%	Detroit, MI	25,147	25,147	0.00%	3,438	-86.33%
Hillsborough County, FL	28,020	28,019	0.00%	11,581	-58.67%	Kansas City, MO	18,283	18,280	-0.02%	4,324	-76.35%
Orange County, FL	27,422	27,434	+0.04%	36,576	+33.38%	Portland, OR	6,430	6,430	0.00%	5,128	-20.25%
Atlanta, GA	37,208	37,080	-0.34%	4,306	-88.43%	Philadelphia, PA	10,417	10,552	+1.30%	4,383	-57.92%
Cedar Rapids, IA*	2,006	2,006	0.00%	264	-86.84%	Aberdeen, SD*	348	348	0.00%	2	-99.43%
Des Moines, IA*	5,236	5,236	0.00%	373	-92.88%	Rapid City, SD*	1,650	1,650	0.00%	0	-100.00%
Boise, ID*	3,426	3,324	-2.98%	1,175	-65.70%	Sioux Falls, SD*	5,735	5,735	0.00%	0	-100.00%
Meridian, ID*	1,916	1,683	-12.16%	945	-50.68%	Austin, TX	12,465	12,571	+0.85%	12,417	-0.39%
Nampa, ID*	2,183	2,060	-5.63%	301	-86.21%	Dallas, TX	35,366	35,326	-0.11%	26,037	-26.38%
Aurora, IL	3,754	3,757	+0.08%	20	-99.47%	Houston, TX	67,045	66,887	-0.24%	28,245	-57.87%
Chicago, IL	98,563	98,570	+0.01%	5,601	-94.32%	San Antonio, TX	38,467	38,852	+1.00%	5,747	-85.06%
Naperville, IL*	1,876	1,879	+0.16%	71	-96.22%	Alexandria, VA*	1,096	1,109	+1.19%	661	-39.69%
Peoria, IL*	3,040	3,082	+1.38%	69	-97.73%	Norfolk, VA*	4,399	4,388	-0.25%	3,077	-30.05%
Rockford, IL*	4,097	4,099	+0.05%	43	-98.95%	Richmond, VA	5,656	5,666	+0.18%	4,008	-29.14%
Springfield, IL*	3,338	3,338	0.00%	102	-96.94%	Virginia Beach, VA*	5,805	5,807	+0.03%	2,050	-64.69%
Overland Park, KS*	3,029	3,029	0.00%	485	-83.99%	Seattle, WA	12,106	12,106	0.00%	3,542	-70.74%
Topeka, KS*	2,997	2,997	0.00%	30	-99.00%	Spokane, WA*	5,762	5,762	0.00%	391	-93.21%

All 40: Official 585,457; USCrash 585,457 (0.00%); US-Accidents 226,150 (-61.37%).

Official references.

AZ ADOT, Arizona Motor Vehicle Crash Facts 2021, city/county table.
CA CHP, 2021 Annual Report, Table 8A–Crashes.
FL FLHSMV, Traffic Crash Facts 2021, county tables (PDF pp. 36–37).
GA GDOT crash dashboard, Atlanta 2021.
ID ITD, Idaho Traffic Crashes 2021, Table 13.
KS KDOT official crash-service city counts, 2021.
SD SD DPS, 2021 Traffic Crash Summary, Table 3-10.
IA Iowa DOT, official crash-service city counts for 2021.
IL IDOT, 2021 City Summaries.
MA MassDOT, official TOWN_SUMMARY service, Boston and Worcester 2021.
MI Michigan Traffic Crash Facts 2021, County/Communities summary (PDF p. 73).
MO MSHP STARS, full-year 2021 area report for Kansas City, city code 1280.
OR Oregon Traffic Crash Summary 2021, Portland city summary.
PA PennDOT, 2021 Pennsylvania Crash Facts and Statistics, Philadelphia County.
TX TxDOT, 2021 Crashes and Injuries by Cities and Towns.
VA Virginia DMV, Virginia Traffic Crash Facts 2021, independent-city totals.
WA Washington State Patrol CAT, Seattle and Spokane full-year 2021 queries.

Across all 40 areas, the official total is 585,457, compared with 585,457 USCrash records (0.00%) and 226,150 US-Accidents records (−61.37%). Figure S1 extends the main-text spatial comparison to Hillsborough County, Atlanta, Dallas, San Antonio, Cedar Rapids, Aurora, Kansas City, and Seattle. Each area uses the same boundary, grid size, and color scale for both datasets. Hillsborough is a county comparison; Aurora and Kansas City refer to Illinois and Missouri, respectively. The maps use retained geolocated crashes within each area, whereas Table S5 follows the source populations described above.

US-Accidents generally shows fewer crash records and sparser spatial coverage. In several areas, its records are concentrated along selected major traffic corridors, while USCrash captures a broader distribution across local streets as well as major corridors. Together with the official-count comparison, these spatial patterns indicate substantial undercoverage of reported crashes in US-Accidents.

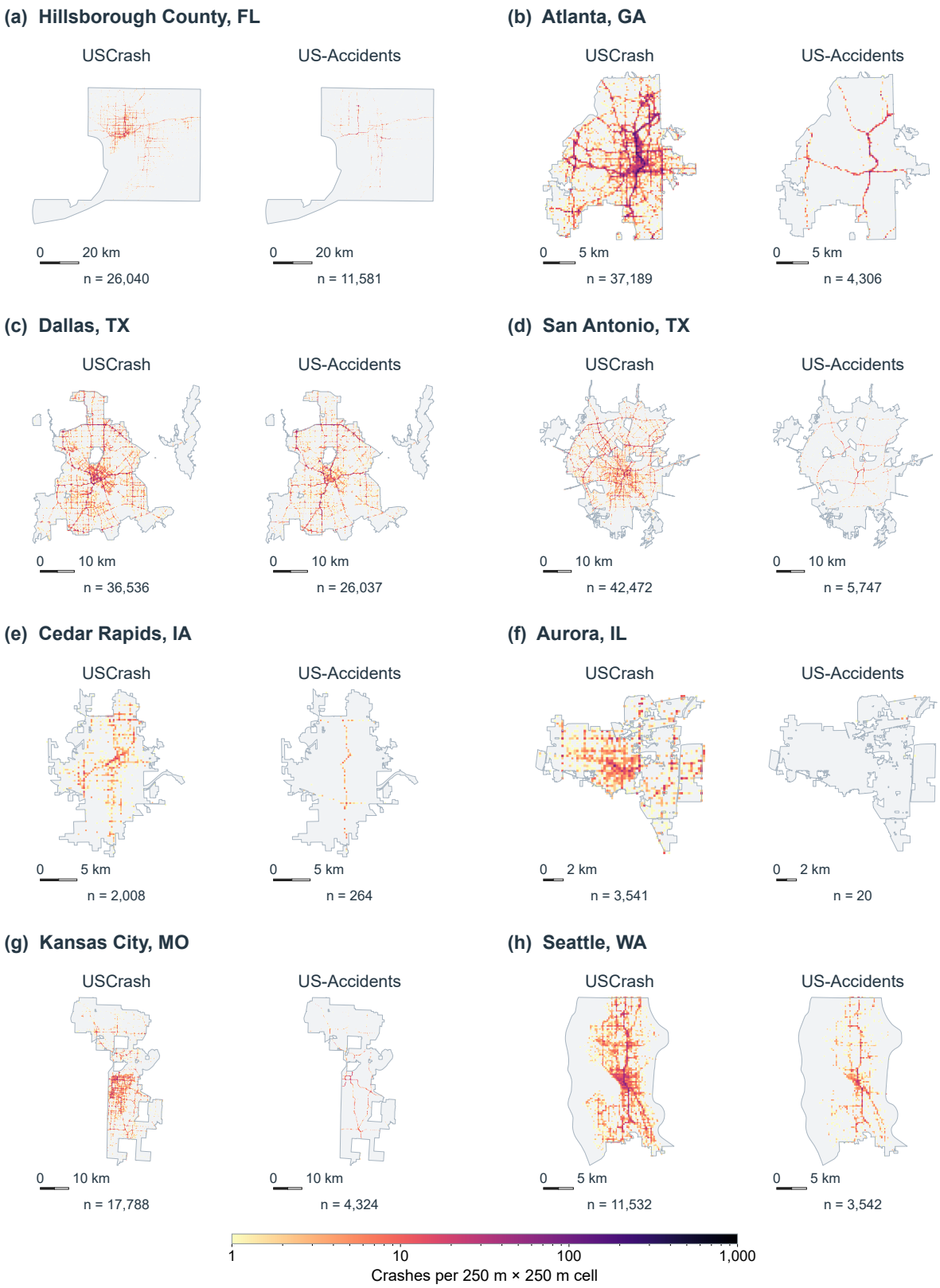


Figure S1: Spatial distributions of geolocated crashes in 2021 for (a) Hillsborough County, FL, (b) Atlanta, GA, (c) Dallas, TX, (d) San Antonio, TX, (e) Cedar Rapids, IA, (f) Aurora, IL, (g) Kansas City, MO, and (h) Seattle, WA. Each area compares USCrash (left) with US-Accidents (right), using identical map extents and 250 m × 250 m cells. All panels use the main-text logarithmic color scale of 1–1,000 crashes per cell. Counts below the maps give the plotted records.

Supplementary Note 5. Data organization and reproducibility

Final crash tables are organized by state and year and use the filename `ST_crashes_harmonized_YYYY.csv`. Original sources, necessary crash-level preparation tables, and final harmonized files remain distinct. Table S6 identifies the resources used to interpret and reproduce the dataset. The code package supplies state/version configurations, category crosswalks, preparation scripts, revision steps, runtime information, and SHA-256 file inventories.

Reproduction follows the source-specific preparation, common-schema harmonization, and applicable enrichment or coordinate-supplementation steps, followed by validation. The representative-year entry point executes this sequence in an isolated workspace. Its adapters select the test year and use preserved donor snapshots for the applicable geographic revisions.

Table S6: Data and code resources. ST and YYYY denote the state code and annual file year; paths and execution instructions are provided by the accompanying code package.

Resource	File(s)	Purpose
Annual crash files	`ST_crashes_harmonized_YYYY.csv`	Final crash-level records, organized by state and year.
Source inventory	`source_inventory_current.csv`	Agency, source URL, retained years, and access notes.
Schema and configurations	`canonical_crash_schema.csv`; `config/states/*.json`	Field definitions and version-specific source bindings.
Category mappings	`code_crosswalk_ST.csv`	Reviewed source-to-common category mappings and provenance.
Input/output catalog	`configured_input_output_links.csv`	Configured source table for each annual output.
Processing code	`harmonize_crash.py`; state-specific builders	Crash-level preparation and semantic harmonization.
Reproduction entry point	`run_representative_replay.py`	Isolated representative-year rerun, including applicable revisions.
Verification records	`verification_results.csv`; `file_manifest.csv`	Recorded reproduction results and retained-file SHA-256 inventory.

A representative-year reproduction was executed for each of the 41 states and DC: 2021 where available and 2019 for New Mexico. All 42 annual outputs were byte-identical to the retained final files and passed structural validation (Table S7). Sixteen cases also reran their state-specific preparation builders; the remaining cases used the preserved configured crash-level inputs. Washington required all ten years from 2017–2026 for cross-year reconciliation, and all ten outputs matched. These tests document the executed representative cases, while the source versions and full annual coverage remain available in the configurations and catalogs.

Table S7: Representative-year reproducibility results. Match denotes byte-identical output; all 42 cases also passed structural validation. Washington's ten-year check is described in the text.

State	Year	Records	Match	State	Year	Records	Match	State	Year	Records	Match
AK	2021	7,924	Yes	KS	2021	54,574	Yes	NY	2021	372,159	Yes
AL	2021	152,592	Yes	KY	2021	131,767	Yes	OH	2021	269,674	Yes
AR	2021	81,037	Yes	MA	2021	125,212	Yes	OK	2021	9,360	Yes
AZ	2021	121,992	Yes	ME	2021	32,646	Yes	OR	2021	45,295	Yes
CA	2021	422,980	Yes	MI	2021	282,640	Yes	PA	2021	118,095	Yes
CO	2021	97,788	Yes	MN	2021	63,791	Yes	SD	2021	19,461	Yes
CT	2021	101,159	Yes	MO	2021	146,757	Yes	TN	2021	236,094	Yes
DC	2021	21,276	Yes	MS	2021	83,319	Yes	TX	2021	633,784	Yes
DE	2021	37,766	Yes	MT	2021	22,924	Yes	UT	2021	61,141	Yes
FL	2021	401,790	Yes	NC	2021	307,117	Yes	VA	2021	118,498	Yes
GA	2021	389,636	Yes	NE	2021	39,706	Yes	VT	2021	10,533	Yes
IA	2021	54,624	Yes	NJ	2021	226,958	Yes	WA	2021	118,124	Yes
ID	2021	27,491	Yes	NM	2019	23,896	Yes	WI	2021	128,296	Yes
IL	2021	295,634	Yes	NV	2021	52,052	Yes	WY	2021	13,903	Yes